

Problem Statement: Optimizing RAG Document Retrieval for Agronomic Advice

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Smallholder farmers increasingly rely on digital advisory tools to get timely answers about crop diseases, pest outbreaks, nutrient deficiencies, soil management, and climate adaptation. Retrieval-Augmented Generation (RAG) systems promise to deliver this advice by grounding generated answers in trusted agricultural extension material rather than relying solely on a language model's internal knowledge, which can be outdated, generic, or regionally inaccurate. However, the quality of a RAG system's output is fundamentally constrained by the quality of its retrieval step: if the documents surfaced in response to a farmer's question are not the most relevant ones available, even a highly capable generation model will produce answers that are incomplete, misleading, or practically unhelpful. This dependency makes the retrieval component not the generator the critical bottleneck for reliable agronomic advice at scale.

Current baseline retrieval methods, such as TF-IDF, rely on simple lexical overlap between a query and candidate documents. This approach struggles with the vocabulary mismatch common in real-world farmer queries, where informal, symptom-based, or colloquial descriptions (e.g., "leaves turning yellow at edges") must be matched against technical extension material (e.g., "potassium deficiency chlorosis"). This limitation reduces retrieval accuracy and, in turn, the usefulness of any downstream generated advice.

This project aims to design and evaluate a retrieval engine that improves upon TF-IDF for matching smallholder farmer questions to relevant documents within a fixed knowledge base of agricultural extension material covering crop diseases, pests, nutrient deficiencies, soil management, and fertiliser guidance. Candidate approaches include BM25, dense embedding-based retrieval, and neural rerankers methods. Performance will be measured using nDCG@5 against expert-annotated relevance judgments, with the goal of maximizing the relevance of the top-5 retrieved documents for a given query.

Improving retrieval performance in this setting has direct practical value: more accurate document retrieval translates into more accurate and trustworthy agronomic advice delivered to farmers who often lack easy access to agricultural extension officers. Because the dataset is

small, clean, and self-contained, and the task requires no GPU infrastructure, this research is feasible to complete within constrained time and compute budgets while still producing results that generalize to real-world agricultural advisory systems.

Describe the problem the selected project aims to solve.

Smallholder farmers often struggle to get accurate, timely answers to questions about crop diseases, pests, nutrient deficiencies, and soil problems. Agricultural extension officers who could normally answer these questions are few, hard to reach, and often serve large areas, so a farmer with a sick crop may wait days or weeks for advice — by which time the crop may already be damaged or lost. Written agricultural guidance exists, but it is scattered across many documents, uses technical language, and is not organized to quickly connect a farmer's specific question to the exact piece of advice they need.

Who is affected by this problem?

- Smallholder farmers, especially those in rural or under-resourced regions with limited access to extension officers, agronomists, or reliable internet-based advisory services.
 - Agricultural extension workers and NGOs, who are stretched thin trying to serve large numbers of farmers with limited staff.
 - Local food systems and rural economies, since crop losses from undiagnosed or late-diagnosed problems affect household income and food security, and can ripple out to communities that depend on that produce.
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Why is solving this problem important?

Left unsolved, delayed or inaccurate agronomic advice leads directly to worse crop outcomes: diseases and pests spread further before being addressed, nutrient deficiencies go uncorrected through a growing season, and farmers may apply the wrong treatment based on guesswork. These outcomes translate into lower yields, greater financial loss for farmers who often operate with very little financial cushion, and reduced food availability at the household and community level. Solving this problem means farmers can get relevant, trustworthy advice at the moment they need it, rather than after damage is already done.

Which values should guide your solution?

- **Accuracy** — retrieved information must genuinely match the farmer's problem, not just superficially resemble it.
 - **Accessibility** — the solution should work for users with plain-language, non-technical questions, not just users who already know the right agronomic terminology.
 - **Trustworthiness** — since the advice may influence real farming decisions, retrieved documents should come from credible, verifiable agricultural sources.
 - **Equity** — the system should perform reliably across a range of question phrasings and topics, not just the ones best represented in the training/evaluation data.
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Why did you choose these values?

These values follow directly from who is affected and what's at stake. Because the end-user is a farmer making real decisions about their crop — not a researcher casually browsing information — accuracy and trustworthiness are non-negotiable: a wrong or irrelevant document could lead to a wrong treatment. Accessibility matters because the whole point of the project is to close a gap for people who don't have easy access to an expert who can translate their plain-language description ("leaves turning yellow") into the right technical category. Equity matters because if the system only works well for well-phrased or common questions, it quietly fails the farmers who need it most — those with less formal agricultural vocabulary or more unusual problems.

What ethical risks should your team consider?

- **Overconfidence in wrong retrievals:** if the system confidently returns a plausible-looking but irrelevant document, a farmer may act on bad advice without realizing it's a mismatch.
 - **Representation gaps:** the knowledge base may cover some crops, regions, or problems far better than others, so farmers with less-represented issues could receive systematically worse results.
 - **Over-reliance / de-skilling:** farmers or local advisors might lean on the tool instead of consulting a human expert even in cases (e.g., unusual disease outbreaks) that genuinely need expert judgment.
 - **Evaluation bias:** since performance is judged against a fixed set of expert relevance judgments, the system could be optimized to match those specific judgments well while still failing on real-world queries that differ in phrasing or intent from the evaluation set.
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How will your team reduce these risks?

- Test retrieval performance specifically on informal, symptom-based, or vaguely worded queries (not just well-formed technical ones), since that's the realistic failure mode for farmers.
 - Check whether performance is uneven across topics (e.g., strong on crop diseases but weak on soil management) and report those gaps honestly rather than only reporting an aggregate score.
 - Keep the retrieval step transparent about *why* a document was retrieved, so a downstream generation step (or a human reviewer) can sanity-check the match rather than treating retrieval as a black box.
 - Avoid overfitting to the specific evaluation set by testing generalization where possible, rather than only tuning until nDCG@5 looks good on the given judgments.
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What outcomes would define success?

- **Quantitative:** a meaningful, consistent improvement in nDCG@5 over the TF-IDF baseline across the full range of query types in the dataset, not just on the easiest cases.
- **Qualitative:** when manually inspecting retrieved documents for realistic, informally-phrased farmer questions, the top results are ones a human agronomist would judge as genuinely useful and relevant — not just keyword-adjacent.
- **Robustness:** the improvement holds across different topic areas (diseases, pests, soil, nutrients, climate) rather than being driven by strong performance in just one category.
- **Practical relevance:** the retrieval approach is reasonably fast and lightweight (no GPU required, as specified), so it could realistically be deployed in a resource-constrained advisory tool rather than remaining a research-only result.